12312515 王络[illegible] Assignment 1

Q1. (a). Simple linear regression model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

Likelihood function: $L = p(y_1, \cdots, y_n) = p(y_1)p(y_2)\cdots p(y_n)$, $y_i \overset{ind}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$

$\Rightarrow l = \sum \log p(y_i) = n\log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2}\sum(y_i - \beta_0 - \beta_1 x_i)^2$ To maximize l:

we need to minimize $S = \sum(y_i - \beta_0 - \beta_1 x_i)^2$
$$\begin{cases} \frac{\partial S}{\partial \beta_0} = -2n(\bar{y} - \beta_0 - \beta_1\bar{x}) = 0 \\ \frac{\partial S}{\partial \beta_1} = -2(\sum x_i y_i - \beta_0 \sum x_i - \beta_1 \sum x_i^2) = 0 \end{cases}$$

Then we can get: $\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})y_i}{\sum(x_i - \bar{x})^2}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$

For σ^2: $\frac{\partial l}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = 0$ $\therefore \hat{\sigma}^2 = \frac{1}{n}\sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$

(b) MLE and LSE have same $\hat{\beta}_0$ and $\hat{\beta}_1$, but for σ^2:

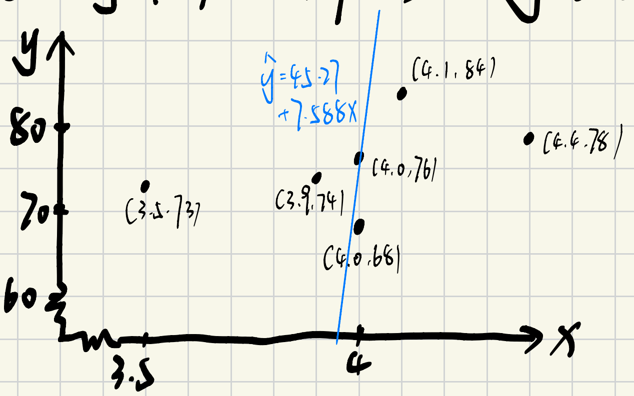
$\hat{\sigma}^2_{LSE} = \frac{n}{n-2}\hat{\sigma}^2_{MLE}$, and $\hat{\sigma}^2_{MLE}$ is biased, $\hat{\sigma}^2_{LSE}$ is unbiased

(c) Let $c_i = \frac{x_i - \bar{x}}{S_{xx}}$, then $\hat{\beta}_1 = \sum c_i y_i$ (Since y_i are independent)

$Cov(\bar{y}, \hat{\beta}_1) = Cov\left(\frac{1}{n}\sum y_i, \sum c_j y_j\right) = \frac{1}{n}\sum\sum c_j Cov(y_i, y_j) \overset{\downarrow}{=} \frac{\sigma^2}{nS_{xx}}\sum(x_i - \bar{x}) = 0$

$\therefore Cov(\bar{y}, \hat{\beta}_1) = 0$ implies they are independent.

Q2 (a)



It seems that there is a positive relationship between x and y but it is not strong

(b)

$\bar{x} = 3.98$ $\bar{y} = 75.5$, $S_{xx} = 0.428$ $S_{xy} = 3.25$ suppose: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

Then we can get: $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = 7.588$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 45.27$

$\therefore$ The least squares line: $\hat{y} = 45.27 + 7.588x$

(c) Linear regression model: $y_i = \beta_0 + \beta_1 x + \varepsilon_i$, where $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

Necessary assumptions: $\mathbb{E}(\varepsilon_i) = 0$, $Var(\varepsilon_i) = \sigma^2$, $Cov(\varepsilon_i, \varepsilon_j) = 0$, $i \neq j$

(d) $H_0: \beta_1 = 0$ $H_1: \beta_1 \neq 0$

$S^2 = \frac{\sum(y_i - \hat{y}_i)^2}{n-2} = 29.71$. $t = \frac{\hat{\beta}_1 - \beta_1}{S/S_{xx}^{\frac{1}{2}}} \xrightarrow{\text{under } H_0} \frac{\hat{\beta}_1}{S/S_{xx}^{\frac{1}{2}}} = 0.911 < t_{0.025,4} = 2.776$

$\therefore$ We can not reject H_0, which means there is insufficient evidence that the duration of an eruption has an effect on the interval to the next eruption.

(e) 95% CI of β_1 : $\hat{\beta}_1 \mp S/S_{xx}^{\frac{1}{2}} \cdot t_{0.025,4} = 7.588 \mp 23.12$

$\therefore$ Required CI: $(-15.532, 30.708)$

(f) $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = 1 - \frac{(n-2)S^2}{143.5} = 0.17$

which means only 17% of the variation in the interval is explained by the duration, Indicating a weak linear relationship.

(g) Prediction: $\hat{y} = 45.27 + 7.588x = 75.63$

Prediction interval : $\hat{y} \pm t_{0.025,4} S\sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{\sum(x_i - \bar{x})^2}} \Rightarrow (59.28, 91.97)$

$\therefore$ The predicted interval to next eruption is 75.63 minutes with a 95% prediction interval of $(59.28, 91.97)$ minutes.